

Caixing Wang

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Working Experience

Southeast University

Aug 2025 – Now

Assistant Professor in School of Statistics and Data Science

The Chinese University of Hong Kong

Aug 2024 – July 2025

Postdoc in Statistics

◦ Mentor: Professor Junhui Wang

Education

Shanghai University of Finance and Economics

Sept 2019 – June 2024

Ph.D. in Statistics

◦ Supervisor: Professor Xingdong Feng, Associate Professor Xin He

Shanghai University of Finance and Economics

Sept 2015 – June 2019

B.S. in Statistics

Research Interests

Statistical Machine Learning; Kernel Methods; Quantile Regression; Large-scale and Distributed Data Analysis; Deep Learning

Journal Publications (* and † refer to corresponding author and equal contributions (or Alphabet ordering))

[1]. *Deep nonparametric quantile regression under covariate shift*. Xingdong Feng[†], Xin He^{*†}, Yuling Jiao[†], Lican Kang^{*†}, **Caixing Wang[†]**. **Journal of Machine Learning Research** 25 (385), 1-50.

[2]. *Communication-efficient nonparametric quantile regression via random features*. **Caixing Wang**, Tao Li, Xinyi Zhang, Xingdong Feng, Xin He^{*}. **Journal of Computational and Graphical Statistics**, 2024, 33(4), 1175–1184.

[3]. *A lack-of-fit test for quantile regression process models*. Xingdong Feng^{*}, Qiaochu Liu, **Caixing Wang**. **Statistics & Probability Letters** 192, 109680, 2023.

Conference Publications

[4]. *Distributed high-dimensional quantile regression: estimation efficiency and support recovery*. **Caixing Wang^{*}**, Ziliang Shen. **International Conference on Machine Learning (Spotlight)**, 2024, 235: 51415-51441.

[5]. *Optimal kernel quantile learning with random features*. **Caixing Wang**, Xingdong Feng^{*}. **International Conference on Machine Learning (Spotlight)**, 2024, 235: 50419-50452.

[6]. *Towards theoretical understanding of learning large-scale dependent data via random features*. Chao Wang, Xin He^{*}, Xin Bing, **Caixing Wang^{*}**. **International Conference on Machine Learning (Spotlight)**, 2024, 235: 50118-50142.

[7]. *Towards a unified analysis of kernel-based methods under covariate shift*. Xingdong Feng[†], Xin He[†], **Caixing Wang^{*†}**, Chao Wang[†], Jingnan Zhang[†]. **Advances in Neural Information Processing Systems**, 2023, 36: 73839-73851.

Preprints

- [8]. *Optimal transfer learning for kernel-based nonparametric regression.* Chao Wang[†], **Caixing Wang[†]**, Xin He, Xingdong Feng. **Major Revision in Journal of American Statistical Association.**
- [9]. *Estimation of conditional extremes in reproducing kernel Hilbert spaces with application to large commercial banks data.* Fang Chen, **Caixing Wang***. **Major Revision in Statistica Sinica.**
- [10]. *Estimation and inference on distributed high-dimensional quantile regression: double-smoothing and debiasing.* **Caixing Wang**, Ziliang Shen, Shaoli Wang, Xingdong Feng. **Under review.**
- [11]. *Distributed learning for adaptive and robust nonparametric regression.* **Caixing Wang.** **Under review.**
- [12]. *Inertial quadratic majorization minimization with application to kernel regularized learning.* Qiang Heng, **Caixing Wang***. **Under review.**
- [13]. *High-dimensional differentially private quantile regression: distributed estimation and statistical inference.* Ziliang Shen, **Caixing Wang**, Shaoli Wang, Yibo Yan. **Under review.**
- [14]. *Improved statistical analysis for spectral algorithms under general conditions: optimality and robustness.* **Caixing Wang**, Qiang Heng, Qi Kuang, Xingdong Feng. **Under review.**
- [15]. *Random projections for spectral algorithms in mis-specified setting: Sobolev norm learning rates and minimax optimality.* **Caixing Wang.** **Under review.**
- [16]. *Generalization properties of robust learning with random features.* **Caixing Wang,** **Under review.**

Reviewer Services

Journal: *Journal of American Statistical Association (JASA)*, *Journal of the Royal Statistical Society Series B (JRSSB)*, *The Annals of Applied Statistics (AOAS)*, *Journal of Computational and Graphical Statistics (JCGS)*, *Statistica Sinica*, *Statistics and Its Interface (SII)*, *Journal of Parallel and Distributed Computing (JPDC)*

Conference: *International Conference on Learning Representations (ICLR)*, *Neural Information Processing Systems (NeurIPS)*

Open-source Software

- **DisRFKQR:** R package for “Communication-efficient nonparametric quantile regression via random features” accepted by JCGS.
- **DQR-covariate-shift:** Python package for “Deep nonparametric quantile regression under covariate shift” accepted by JMLR.
- **DHSQR:** R package for “Distributed high-dimensional quantile regression: estimation efficiency and support recovery” accepted by ICML.
- **Kernel-CS:** Python package for “Towards a unified analysis of kernel-based methods under covariate shift” accepted by NeurIPS.

Talks

International Conference on Statistics, Data Science and Artificial Intelligence (2024)

CCUT, Changchun

- Deep quantile regression under covariate shift

The 1st International Conference for PhD Pioneers (2024)

RUC, Beijing

- Optimal transfer learning for kernel-based nonparametric regression

The 2nd Conference for Chinese Statistical Association of Young Scholars (2024)

JNU, Xuzhou

- Towards a unified analysis of kernel-based methods under covariate shift

Teaching Experience

Teaching Assistant

2020-2024

- Undergraduate Courses: Survival analysis, Extreme value theory.

Skills

- **Language:** Strong reading, writing and speaking competencies for English and Chinese.
- **Programming:** Python, R, C++.
- **Operating Systems:** Windows, MacOS, Linux.
- **Documentation:** LATEX, Markdown, MS Office.